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Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	

Title of the Project: Multimodal Fusion Network for Chest X-ray Pneumonia Detection Using Deep Learning

Abstract

Pneumonia is a serious respiratory disease that causes significant illness and mortality worldwide, particularly among children, older adults, and individuals with weakened immune systems. Accurate and timely diagnosis is essential for effective treatment. Chest X-ray imaging is one of the most commonly used diagnostic methods; however, many existing deep learning systems rely solely on X-ray images, whereas clinicians also consider patient information such as age, body temperature, oxygen saturation (SpO_2), and symptoms before making a diagnosis. This limitation motivates the need for a multimodal approach that combines both visual and clinical information.

The primary stakeholders of this project are hospitals, diagnostic centers, healthcare professionals, and patients who require timely pneumonia screening. Existing AI-based solutions often focus only on image classification, which may not fully capture the patient's clinical condition. Studies in multimodal deep learning indicate that integrating medical images with structured clinical data can improve diagnostic performance and produce more reliable predictions.

This project proposes a Multimodal Fusion Network for pneumonia detection by combining chest X-ray images with clinical patient information. The chest X-ray image is processed using a Convolutional Neural Network (CNN) to extract visual features, while structured clinical parameters such as age, temperature, SpO_2, fever, and cough are processed using a dense neural network. The extracted feature representations are fused into a unified feature vector, which is then used for binary classification into Normal or Pneumonia. The proposed model will be evaluated against image-only and clinical-data-only baseline models using metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix to assess the effectiveness of multimodal fusion.

The scope of this project is limited to binary pneumonia classification using chest X-ray images and selected clinical parameters. It does not include disease severity estimation, treatment recommendation, or integration with hospital information systems. The expected outcome is a research-oriented prototype that demonstrates improved diagnostic performance over conventional image-only models while highlighting the benefits of multimodal fusion for computer-aided pneumonia detection.

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References (Category specific)

1. C. Hsieh *et al*., "MDF-Net for abnormality detection by fusing X-rays with clinical data," *Scientific Reports*, vol. 13, Art. no. 15873, 2023.

2. J. Hofmeister *et al*., "Validating the accuracy of deep learning for the diagnosis of pneumonia on chest X-ray against a robust multimodal reference diagnosis," *European Radiology Experimental*, vol. 8, no. 1, 2024.

3. S. Singh *et al*., "Efficient pneumonia detection using Vision Transformers on chest X-rays," *Scientific Reports*, vol. 14, 2024.

4. D. Li, "Attention-enhanced architecture for improved pneumonia detection in chest X-ray images," *BMC Medical Imaging*, vol. 24, 2024.

5. S. Shurrab and F. E. Shamout, "Multimodal masked siamese network improves chest X-ray representation learning," *Scientific Reports*, vol. 14, 2024.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A6143	GUTTA SANDEEP	
2	23BQ1A6146	JASTI LOHTHA	
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Project Guide

Name: Signature:

MAJOR PROJECT REGISTRATION FORM

Date:

INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. All of you must chose '**Research oriented**' projects only. Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the '**Project category**' section. Also provide the link to the problem statement in the '**References**' section.

To write Abstract

(Use the following as a mental checklist)

a) **Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."

b) **Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."

c) **Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.

d) **Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)

e) **Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.

f) **Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?

g) **Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.

Project Coordinator HOD-CSE (AI)